Awais Asghar

AI and Machine Learning Engineer

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Core Areas

Computer Vision (CV), Deep Learning (DL), LLMs and RAG, Fine-Tuning and Evaluation, Edge AI and Model Compression, Scientific ML

Education

National University of Sciences and Technology (NUST) — Islamabad, Pakistan

Bachelor of Engineering in Electrical Engineering | *Degree* | *Transcript* — *Sept 2022 - June 2026*

- CGPA: 3.63/4.0 Specialization GPA: 3.94/4.0 Merit Scholarship, 2024 - 2026
- Thesis: ML-Based Hardware Accelerator for Real-Time Image Segmentation on FPGA
- Coursework: Machine Learning, Deep Learning, Computer Vision, Embedded System Design, Microprocessor Systems, Digital System Design, Parallel and Distributed Processing, Digital Signal Processing, Probability and Statistics

Experience

Mitacs Globalink Research Intern — June 2026 - Oct 2026

NRC Herzberg Astronomy and Astrophysics, with Dr. Sebastien Fabbro | *Award Letter* — *Victoria, BC, Canada*

- Built a DL system (PyTorch) that detects faint, icy objects in the outer solar system by spotting slow motion between telescope images taken hours apart, while stars stay fixed
- Processed the CFHT CLASSY survey (about 29 TB of images) on the CANFAR GPU cloud platform; trained on synthetic objects injected into real images so the model learned from known answers
- Raised result reliability: 1,000 of 1,015 reported objects were real (98.5% precision), up from 81%
- Built a fully automatic search, with no human checking, that finds half of all objects down to magnitude 26 (about 100 million times fainter than the faintest star visible by eye), close to the survey team's manually verified 26.25

Deep Learning Research Intern — June 2024 - Aug 2024

Deep Learning Lab (DLL), NUST SEECS | *Certificate* — *Islamabad, Pakistan*

- Completed a 12-week ML and DL program, then trained and benchmarked CNN and transfer learning models in PyTorch and TensorFlow using CUDA on GPUs

Machine Learning Research Intern — June 2024 - Aug 2024

HamsanTech | *Certificate* — *Islamabad, Pakistan*

- Built a skin cancer classification pipeline (91% accuracy, 0.963 AUC) using an ensemble of XGBoost, AdaBoost, LightGBM, SVM, and Logistic Regression
- Developed Power BI and Tableau dashboards and cleaned and analyzed real industry datasets

Startup

CashFlow: Receipt Ingestion and Money Tracker | *Founder* | *Next.js, React, TypeScript, OCR* | Live Demo

- Founded and built a money tracker web app (MVP) that reads receipt photos with OCR, extracts structured transactions, and keeps borrowed and lent money separate from income; deployed on Vercel as a solo founder

Selected Projects

ML-Based Hardware Accelerator for Image Segmentation | *Python, PyTorch, Vitis HLS, Vivado*

- Achieved 56x model compression to 570 KB with 97% pixel accuracy at 2.72 W power on a Zybo Z7-20 FPGA, as my Senior Design Thesis, covering the full pipeline from model training and quantization to hardware design in Vitis HLS and Vivado and real-time edge deployment

Lightweight U-Net for Driving Scene Segmentation | *Python, PyTorch, OpenCV* | GitHub

- Designed a lightweight U-Net for driving scene segmentation that cuts the model from 8.7M to 0.53M parameters while holding 72% IoU, 80% F1, and 96% test accuracy on the CARLA simulator dataset, built in PyTorch and OpenCV; first manuscript draft complete

SureSend: Scam-Safe Payments (PayPal AI Hackathon) | *Next.js, TypeScript, Postgres, Groq, PayPal API* | GitHub | Live Demo

- Built a payment-safety app for scams where victims send the money themselves: it holds the PayPal payment, an AI agent reviews the scam message or voicemail, and a named family member must confirm before the payment is released or voided
- Added fixed red-flag rules that set a minimum risk level the AI can raise but never lower, with no direct AI access to payments; deployed on Render with Postgres and a scheduled job

AI-Based Exam Generator (RAG) | *Python, LangChain, FAISS, Groq API, Llama 3.1* | GitHub

- Built an end-to-end RAG pipeline that turns lecture slides into structured exam papers, covering document ingestion, semantic retrieval with FAISS vector search, and LLM-based question generation with Llama 3.1 (Groq API) via LangChain

TechStore Support Chatbot | *JavaScript, Node.js, Vercel Serverless, Groq API* | GitHub

- Building an English and Roman Urdu support chatbot for an online electronics store: a web widget calling Node.js serverless APIs on Vercel, backed by Llama 3.3 70B on Groq (in progress)

Real-Time Fabric Defect Detection | *Python, OpenCV, NumPy, Jetson Nano* | GitHub

- Built a classical CV pipeline (GLCM, FFT, Gabor filters, IoU-based fusion) for interpretable, low-power fabric defect localization that runs in real time on a Jetson Nano with no deep learning required; first manuscript draft complete

SkinSense: Multi-Model Skin Cancer Classifier | *Scikit-learn, XGBoost, PyTorch* | GitHub

- Built an ensemble classification pipeline (XGBoost, AdaBoost, LightGBM, SVM, Logistic Regression) reaching 91% accuracy and 0.963 AUC for single-image RGB skin cancer detection

Technical Skills

ML & AI: Python, C/C++, PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, NumPy, Pandas, OpenCV, SciPy, CUDA
LLMs & GenAI: LLMs, Prompt Engineering, Function Calling, Structured Output (JSON), RAG, Advanced RAG (HyDE, Multi-Query, CRAG, Self-RAG), Hybrid Search, Reranking
Fine-Tuning: LoRA, QLoRA, PEFT, Embedding Fine-Tuning, Contrastive Learning, Unsloth, Hugging Face Transformers and Hub
Frameworks: LangChain (LCEL), LangGraph, CrewAI, Sentence-Transformers, RAGAS, DeepEval
Vector Search: Qdrant, FAISS, Embeddings, Semantic Search, BM25, Chunking Strategies
Evaluation & Observability: RAG Evaluation, LLM-as-a-Judge, LangFuse, Tracing, Latency and Cost Monitoring
Backend & Web: FastAPI, REST APIs, Pydantic, Async Programming, pytest, Node.js, TypeScript, JavaScript, Next.js, React
Databases: Supabase, PostgreSQL, MySQL, SQL Server
MLOps & DevOps: CI/CD, GitHub Actions, Docker, GitHub Container Registry, Secrets Management, Model Deployment, Hugging Face Spaces, Vercel, Render
Edge AI & Hardware: FPGA (Vitis HLS, Vivado), Jetson Nano, Quantization, Model Compression
Data & Tools: HDF5, CANFAR, Power BI, Tableau, Git/GitHub, Linux, Jupyter, LaTeX

Research Papers (First Drafts Complete)

Design of a Novel Lightweight U-Net Architecture for Efficient Semantic Segmentation: 8.7M to 0.53M parameters, 72% IoU and 96% test accuracy on CARLA
Real-Time Fabric Defect Detection on Edge Devices Using Classical Computer Vision: multi-method pipeline deployed on Jetson Nano

Certifications

Machine Learning Specialization (3 courses), DeepLearning.AI
Deep Learning Specialization (5 courses), DeepLearning.AI
Introduction to Deep Learning (6.S191), MIT

Honors and Achievements

- Millennium Fellow: selected among the top 4% globally by the UN Academic Impact and Millennium Campus Network
- NUST SEECS Merit Scholarship (2024 - 2026) and Dean's Honor List, recognized 4 times
- Prime Minister's Laptop Scheme: awarded a laptop for academic merit
- 3rd position in the Multan Board (BISE Multan) in FSc among 70,000+ students